Dimensionality Reduction for Sparse Movie Ratings Data: PCA via Efficient Power Method Implementation

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Code Repository

1 Research Objective

The primary objective of this research is to efficiently extract principal components from a high-dimensional, strictly sparse movie ratings matrix. Standard dimensionality reduction algorithms become computationally prohibitive when explicitly mean-centering large, sparse datasets, as the process densifies the matrix. To resolve this, the core methodological focus of this project is the manual development of a Simultaneous Power Method algorithm, utilizing an "implicit centering" algebraic modification to preserve matrix sparsity while iteratively computing the dominant eigenvectors and eigenvalues.

Building upon this goal, the other objective is to apply Principal Component Analysis (PCA) to recover latent archetypes determining movie ratings. Finally, the project also investigates how Age and Gender are associated with a user's location within this latent space.

2 Data Description and Matrix Properties

The MovieLens 100k dataset serves as a benchmark for evaluating latent factor models in recommendation systems. It consists of 100,000 ratings (ranging in ordinal values from 1 to 5) provided by 943 users across 1,682 movies.

2.1 Matrix Sparsity and Computational Constraints

The movie ratings data matrix $X \in \mathbb{R}^{943\times1682}$ is characterized by 93.7% sparsity. As shown in the Spy Plot (Figure 1), the observed ratings (black dots) are few compared to missing values (white space). Explicitly centering this matrix would convert it into a dense format, potentially crashing R (RStudio, for example) when run on standard consumer hardware during operations such as multiplication of large, dense matrices.

Table 1: Data Matrix Description

Metric	Value
Total Number of Cells	1,586,126
Actual Ratings	100,000
Missing Ratings	1,486,126
Sparsity Percentage	93.7%
Density Percentage	6.3%

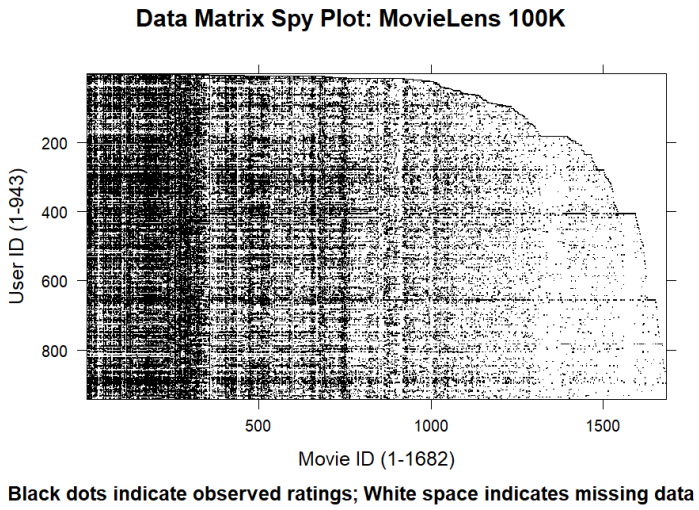

Figure 1: Spy plot of the data matrix highlighting extreme sparsity.

2.2 Genre Distribution

A preliminary analysis of the distribution of movies by genre in the data (Figure 2) reveals a high density in Drama (725 movies) and Comedy (505 movies). These categories provide the dominant signals expected in the primary principal components.

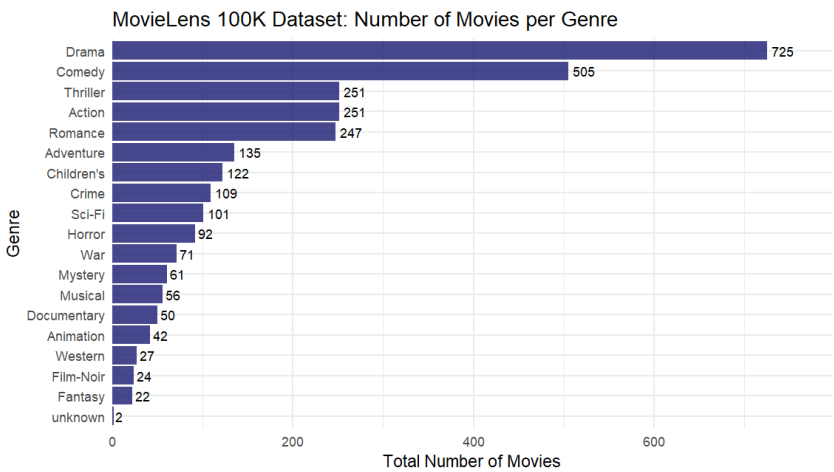

Figure 2: Distribution of movies by genre in the MovieLens 100K dataset.

3 Methodology

3.1 PCA via SVD

Principal Component Analysis on the movie ratings data matrix by approximating the centered ratings matrix X using Singular Value Decomposition (SVD):

$$X \approx U\Sigma V^T$$

1. Movie Loadings (V): An orthonormal matrix $V \in \mathbb{R}^{p\times k}$ where each row represents a movie's coordinates in the k-dimensional latent space.

- These loadings define the orientation of the principal axes, allowing us to interpret what movie features a principal axis may correspond to.
- In the loadings plots, the (x, y) coordinates for a movie are simply its values in the corresponding columns of V.

2. PC Scores ($U\Sigma$): The coordinates assigned to users in the space spanned by principal components.

- $U \in \mathbb{R}^{n\times k}$ provides the orthonormal basis for user preferences.
- By scaling U by the singular values (Σ), we obtain the projections of users onto the principal component axes. These "PC scores" give an idea of the "importance" (weightage) a user places on a principal component axis while rating.

3.2 Implicit Centering and Power Method

To preserve matrix sparsity and reduce computational costs, centering on the data matrix is performed "implicitly" (by splitting an iteration into two steps) in the Simultaneous Power Method. Let μ be the vector of column means of the data matrix X. Then, the two steps performed are:

- **Forward Pass:** $U_{tmp} = XQ - \mathbf{1}(\mu^T Q)$
- **Backward Pass:** $V_{raw} = X^T U_{tmp} - \mu(\mathbf{1}^T U_{tmp})$

The modified Power Method algorithm incorporating these steps can be written as:

Algorithm 1 Simultaneous Power Method with "Implicit Centering"

```
Initialize V^(0) ∈ R^{p×k} with random Gaussian entries.
Q^(0) = GramSchmidt(V^(0))
while not converged do
    U_tmp = X Q^(m-1) − 1(μ^T Q^(m-1))
    V_raw^(m) = X^T U_tmp − μ(1^T U_tmp)
    Check convergence: Σ_{i=1}^{k} ||v_{raw,i}^(m) − v_{raw,i}^(m-1)|| < ε
    Q^(m) = GramSchmidt(V_raw^(m))
end while
Recover Singular Values: σ_i^2 = ||v_{raw,i}^(m)|| for i = 1, ..., k
Recover Right Singular Vectors: Column vectors of V = Q^(m)
Recover Left Singular Vectors: u_i = X̃ q_i / σ_i
```

4 Results and Interpretation

4.1 Key Metrics for diagnosis

The convergence and reliability of the recovered outputs from the Power Method are assessed through two primary metrics:

- **Log-Linear Convergence:** The error in the i-th eigenvector, defined as $E_m = \|v_i^{(m)} - v_i^{(m-1)}\|$, follows a geometric decay such that $\log(E_m) \propto -m$, appearing as a straight line on a log-linear plot.

- **Proportion of Variance Captured:** The importance of each principal component is quantified using the squared singular values σ_i^2 relative to the total variation in the centered ratings matrix:

$$PV_i = \frac{\sigma_i^2}{\sum_{j=1}^{p} \sigma_j^2} = \frac{\sigma_i^2}{\|X_{centered}\|_F^2} \tag{1}$$

  where $\|X\|_F$ denotes the Frobenius norm.

In this analysis, the Power Method algorithm achieved the expected log-linear pattern of convergence over 649 iterations. The chosen error threshold was $\epsilon = 10^{-6}$ (Figure 3). The Scree Plot (Figure 4) confirms that the first 10 PCs capture over 35% (35.28% to be exact) of total variation in data. Further, the proportion of variance explained decreases much slowly after PC2 and even slowly after PC4, resulting in elbowing in the graph at those points. This hueristically may lead us to analyse the first two (or broadly, the first 4) principal components for our analysis. This has been done elaborately in the next section.

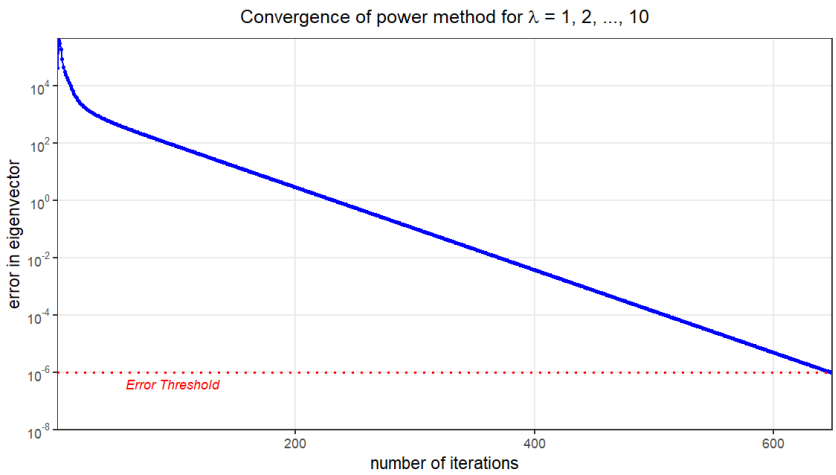

Figure 3: Geometric decay of eigenvector error over iterations.

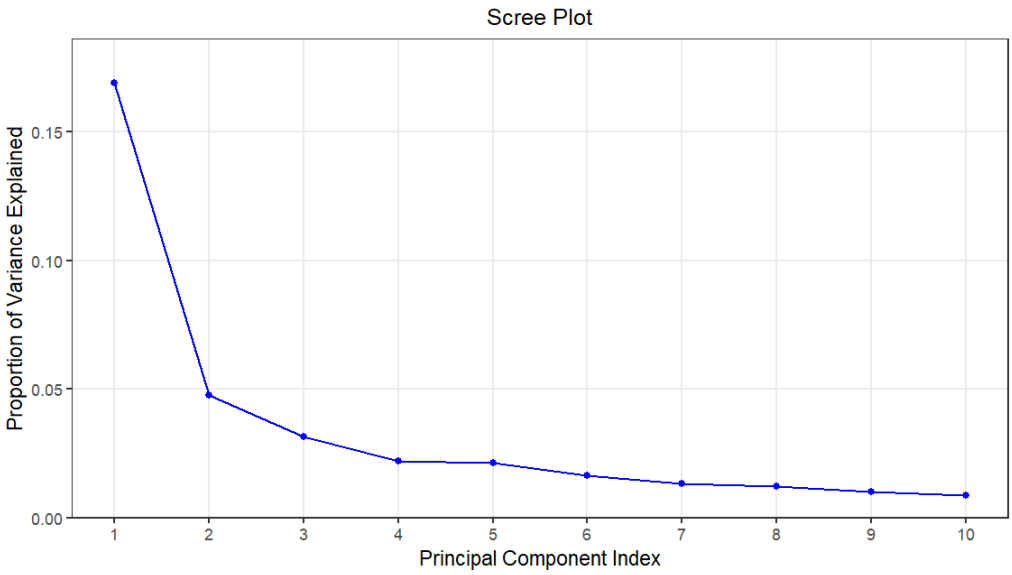

Figure 4: Scree plot of variance captured by the first 10 PCs.

4.2 Interpretation of Loadings (PC1 & PC2)

The loadings plot (Figure 5) gives a visual idea of the latent features influencing weights assigned to movies on PC1 and PC2.

(**Click on the interactive PCA loadings plot linked here** to hover over points to reveal the name, full genre profile, and year of release for each movie. Further, click on the genre keys in the legend to isolate dots representing movies assigned to particular genres.)

- **PC1 (Market Engagement):** This axis may be interpreted as a proxy for viewership and cultural saturation. High positive loadings are dominated by ubiquitously recognized movies such as *Star Wars* and *Fargo*, representing movies that transcend niche demographics to achieve broad market penetration.
- **PC2 (Critical Acclaim vs Blockbusters):** This axis represents a fundamental divergence in cinematic taste, separating high-brow critic favourite films (dominated by the drama genre) from popular blockbusters (dominated by the action genre).
  - **Top 5 Loadings (Cerebral/Drama Movies):** In Figure 6, movies such as *Annie Hall*, *Citizen Kane*, and *Dr. Strangelove* cluster at this extreme. These films often require higher cognitive engagement and are associated with film critics' likings.
  - **Bottom 5 Loadings (Visceral/Action Movies):** This space is occupied by high-octane, VFX-heavy films like *Independence Day*, *The Rock*, and *Twister*. These titles prioritize spectacle and broad-base entertainment value over complex narrative structures.

Figure 6 further illuminates discussion on what PC3 and PC4 may represent. For PC3, highest loadings are dominated by classic action films, whereas lowest loadings are dominated by drama films. Similar genre segmentation can be seen for PC4 loadings.

However, these interpretations must be taken with a grain of salt as in this analysis, only one genre has been assigned to each movie. Hence, PC1 and PC2 have much clearer interpretations as compared to PC3, PC4, and so on.

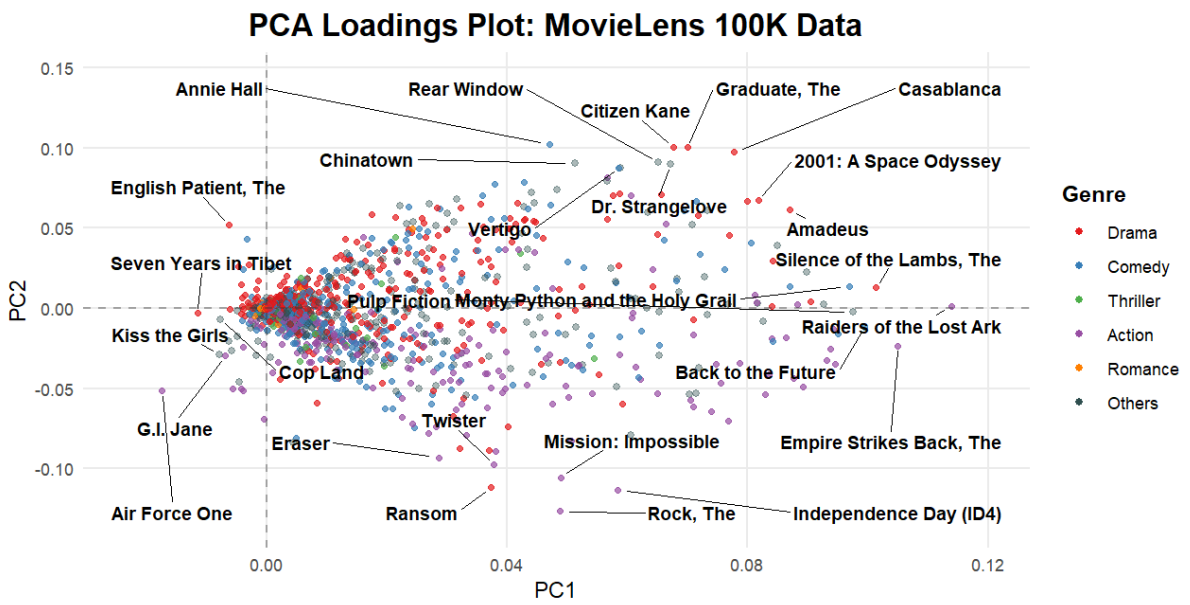

Figure 5: PCA Loadings Plot showing the PC1-PC2 subspace and clustering of movies by loadings.

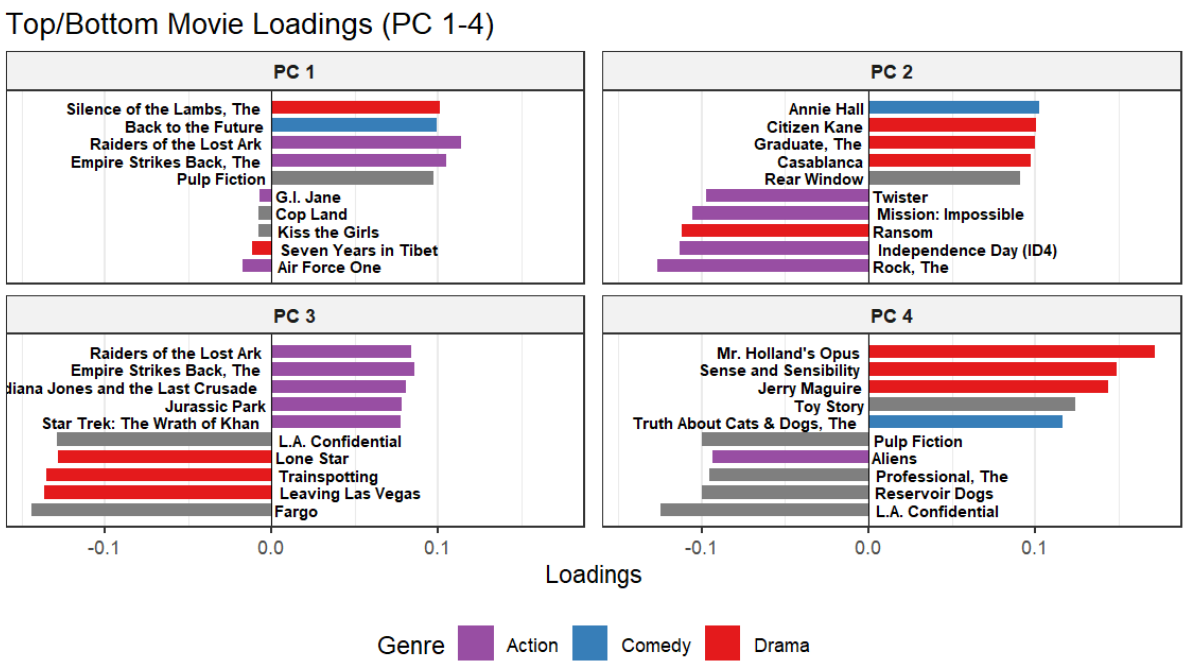

Figure 6: Top 5 and Bottom 5 movie loadings for PCs 1-4.

(**Note on Genre Assignment in Plots:** For colour coding and labelling movie genres, each movie was assigned the first genre listed in its genre profiles in the raw data.

Movies matched to genres other than the top 5 most frequently appearing ones (from Figure 2) have been cubbed under the "Others" label. This makes it easier to visualise segmentation by genre, but may conceal information gained from analysing each film's full genre profile.)

5 Demographic Inferences

In the context of this analysis, the **PC Scores** (represented by the matrix $U\Sigma$) allow us to map each of the 943 users' specific preference towards archetypes represented by each of the 10 principal component axe.

5.1 Gender Divergence

The PC Scores plot (Figure 7) highlights a divergence in centroids (defined as the coordinated of PC1 Scores and PC2 scores for Male and Female users respectively) along PC1. This suggests that genre is an important predictor of how users rate movies along the Market Engagement Axis. To determine if this difference in mean PC2 scores between men and women is significant, the following t tests were conducted:

Table 2: Two-Tailed t-test Results for Gender-Based Mean Differences in PC Scores

Component	**t-statistic**	**p-value**	**Significance**
PC1 (Market Engagement)	-3.4422	0.0006201	Significant at 1% level
PC2 (Critical Acclaim vs Blockbusters)	0.40235	0.6876	Not Significant

The results suggest that the main (significant) demographic division between male and female viewership is strictly a matter of engagement volume and mainstream saturation (measured by PC1). Male and female users don't significantly differ on the importance (average PC scores) they place on the Critically acclaimed vs. Blockbuster films axis (PC2).

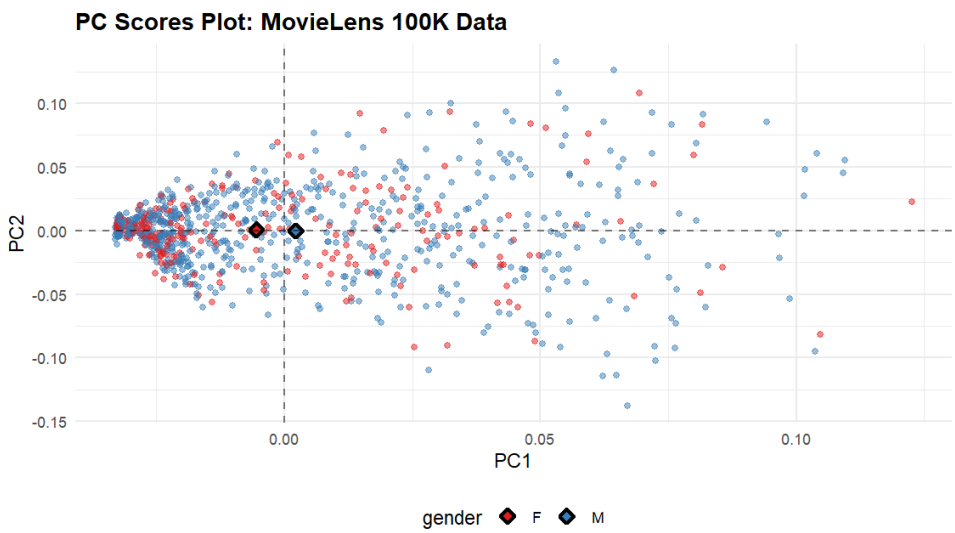

Figure 7: Users projected in the PC1-PC2 space, indicating PC scores.

5.2 Age and Preference Evolution

Correlation analysis (Figure 8) shows that while PC1 scores are age-neutral (corr: -0.08), PC2 scores exhibit a positive (not too strong) correlation of 0.369 with users' ages. This indicates a systematic shift of users' preferences towards the "critically acclaimed" cluster of movies with growing age.

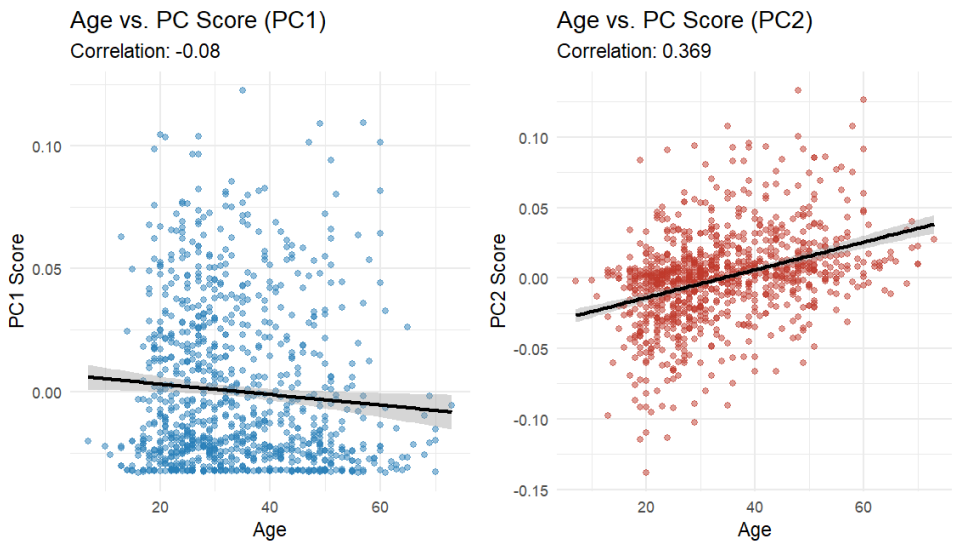

Figure 8: Correlation between Age and PC scores showing preference evolution.

6 Additional Analysis: Gender-Stratified PCA

6.1 Task and Motivation

To further investigate the intersection of demographics and cinematic preference, an additional analysis was conducted by partitioning the sparse ratings matrix into two distinct submatrices: one comprising solely female users (X_{female}) and the other comprising only male users (X_{male}). The Simultaneous Power Method was independently applied to both matrices.

The primary motivation behind this stratification was to determine whether the latent "archetypes" defined by the principal components themselves differ inherently by users' gender, or if the interpretation for principal component axes remains the same as earlier in both cases.

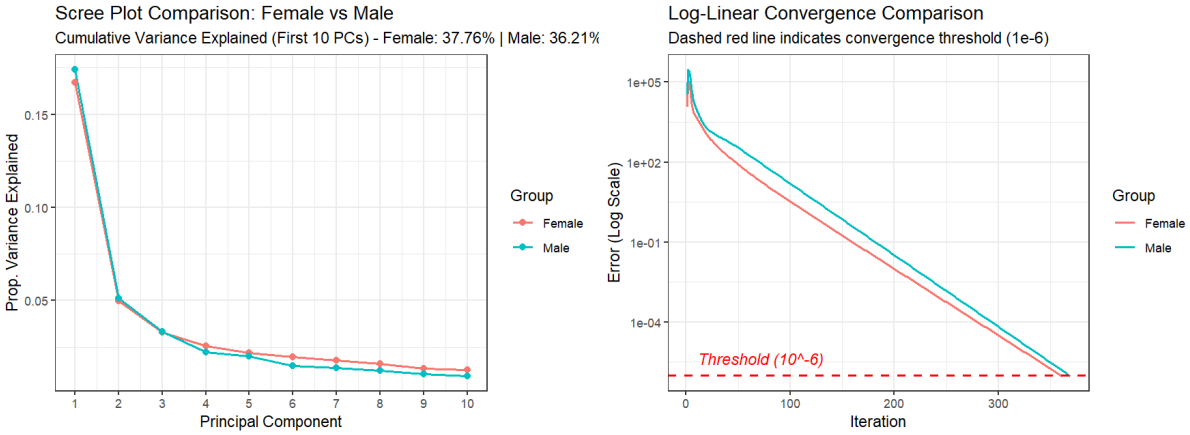

Figure 9: Comparison of variance captured and convergence rates after power method implementation on both genders' submatrices.

6.2 Interpretation of Diagnostic Metrics

The two plots (Figure 9) reveal that the power method convergence and recovered metrics were similar in both cases:

- **Scree Plot Alignment:** The curves tracking proportion of variance explained are nearly identical, with the first ten components capturing 37.76% of the variance for females and 36.21% for males.
- **Convergence Rates:** Both models exhibit nearly parallel log-linear error decay, cleanly hitting the 10^{-6} threshold between 350 and 400 iterations. This shows that the spectral gap (the ratio between successive eigenvalues) is consistent across genders, allowing the Power Method algorithm to perform efficiently on both submatrices.

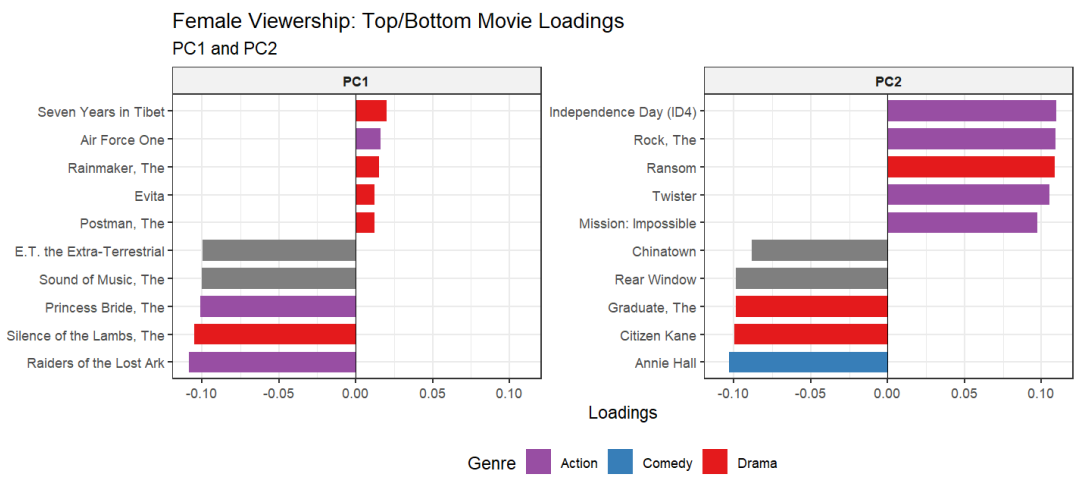

Figure 10: Top and Bottom movie loadings for Female Viewership (PC1 and PC2).

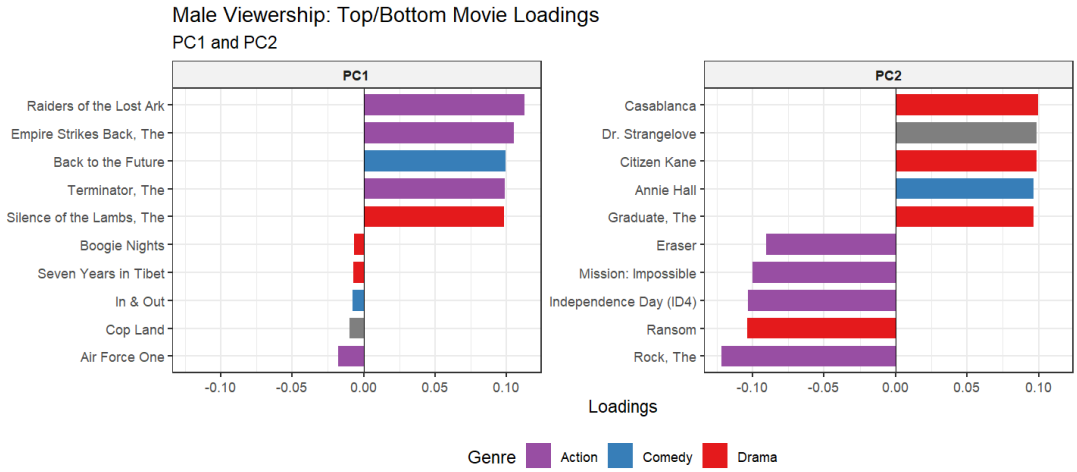

Figure 11: Top and Bottom movie loadings for Male Viewership (PC1 and PC2).

6.3 Interpretation of Loadings and Stability of PCs

An examination of the loadings plots (Figures 10 and 11) demonstrates that the qualitative interpretation of the principal components is quite stable across genders, despite some mathematical artifacts.

- **PC1 (Market Engagement):** For both males and females, the first component is defined by culturally ubiquitous, high-engagement films (e.g., *Raiders of the Lost Ark*, *Silence of the Lambs*).
- **PC2 (Critical Acclaim vs Blockbusters):** The second component identically identifies the divergence between 1990s Action/Blockbusters (*The Rock*, *Independence Day*) and Critical Classics (*Citizen Kane*, *Annie Hall*).

Note on Signs of Loadings: It is crucial to observe that the poles for both PC1 and PC2 are flipped between the male and female plots. For instance, in the female viewership plot, Action films hold positive loadings on PC2, whereas in the male model, they hold negative loadings. Because recovered singular vectors are only unique up to a sign, meaning v and $-v$ span the same 1D subspace, this flip is a purely arbitrary mathematical artifact of the initialization step in the Power Method, not a reversal of actual taste in movies along the axes.

6.4 Additional Analysis Conclusion: Loadings vs. Scores

While this stratified analysis successfully confirms that the interpretation for PC1 and PC2 axes seems to exist independently within both male and female populations, it highlights a distinct methodological challenge. Directly comparing the principal components (V_{female} vs V_{male}) is difficult due to indeterminancy in whether the sign of loadings will be flipped.

The approach utilized earlier in this report—computing a single analysis on the full data matrix and subsequently testing for significant differences in the **PC Scores** ($U\Sigma$) between centroids of the two genders' scores, provides a more interpretable and sound method for quantifying how users differ along their movie preferences based on gender.

7 Conclusion

This project successfully demonstrated the application of dimensionality reduction on highly sparse (93.7%) user-item interaction data. By implementing a modified Simultaneous Power Method equipped with "implicit" centering, the computational bottlenecks caused by operations on dense matrices were entirely bypassed. This algorithm efficiently recovered a low-rank subspace, with the first ten principal components capturing 35.28% of the total market variance.

By successfully distilling sparse ratings data into interpretable axes, the findings suggest that movie recommendation systems can optimize predictions by leveraging these latent archetypes (PCs) along which users' ratings differ.